

Fossil Fuel Freezeout: Iceland as a Model for Post-Carbon Survival

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Introduction

Stephen M. Gardiner (2011), a philosophy professor at the University of Washington, describes climate change as "a perfect moral storm." Billions of individuals and hundreds of nations contribute to the problem, and greenhouse gases diffuse across the globe in such a way that the harm is not contained and assigning blame is extremely difficult. Additionally, the worst effects of today's emissions will not be felt until decades or even centuries into the future. The present generation reaps the benefits of cheap fossil fuels, while the unborn generations, unable to negotiate, hold us accountable, or retaliate, are left with our willful ignorance.

There is no single global institution strong enough to enforce mutual coercion, and without such, a tragedy of the commons persists where every country's short-term self-interest is to pollute, even though it leads to a collective, long-term ruin. It seems that the only way to save the planet is for individual nations to step up one at a time and become ethical martyrs that move towards sustainability by pooling their finite resources into solving a long-term problem that requires complete collaboration to succeed.

Yet current solutions lack long-term efficacy. While afforestation/reforestation of trees sequesters CO₂ at very high rates (Robinson et al., 2025), trees lack the permanence needed to keep up with the globe's emissions. The carbon taken in by these trees is not stored away forever. Carbon stored in trees can be released through wildfires, pests and diseases, drought, and storms, which all become increasingly more common due to climate change itself (NASA Science Editorial Team, 2022). Even naturally dying trees expel carbon when they are decomposing (Government of Canada, 2025). Moreover, if all suitable land in the world were used to plant trees – 0.9 billion hectares that could store 205 gigatons of carbon (Bastin et al., 2019) – the trees would only store 20 years of our current annual emissions in their entire lifetime.¹

Peatland restoration is another common method of capturing carbon. Even though peatlands only cover 3% of the terrestrial globe, they store twice as much carbon as all the world's forests combined (Masterson & North, 2025). However, as carbon sinks, peatlands share similar vulnerabilities to afforestation. Emissions from damaged peatlands account for 5% of all global anthropogenic greenhouse gas emissions, and they can be more easily damaged by drought and fire than trees can (IUCN Team, 2021).

Afforestation/reforestation and peatland restoration are the most common methods of carbon sequestration, sequestering more than 99.9% of the 2 GtCO₂ currently captured. Less than 0.1% of total CO₂ capture comes from novel methods like direct air capture (DAC) (Geden et al., 2024). These novel methods typically fall short because the concentration of CO₂ in the air is so diffused – about 420 to 430 parts ppm (Fajardy et al., 2025). This makes methods like DAC extremely energy intensive and expensive to the point where only rich nations with virtually free energy – such as Iceland and Norway - can successfully implement them without polluting more than they take in to operate equipment.

The next most logical solution to turn to is reducing our own emissions, but such reductions demand personal sacrifice—the least marketable solution in a consumption-driven world. Instead, solutions for renewably maintaining current usage are commercially touted by capitalistic companies boasting sustainability through mitigating our second biggest emitter, transport, which emits 8.14 billion tons of greenhouse gases a year (Ritchie et al., 2022). Yet these solutions are not as clean as people think. Hybrid vehicles pollute almost as much as petrol cars (Niranjan, 2025). In my interview with Iceland's Head of the Environment Agency of Iceland, Arni Finnson, he was quick to point out that “I think the hybrids... they're just fake. That's a greenwashing thing... The government made this compromise with the car importing industry back in 2018 or 2019. It's just fake.” Mr. Arni Finnson is referring to a low emission excise duty loophole vehicle companies use by taking advantage of a law in Iceland that exempts vehicles emitting up to 80 CO₂/km from import excise duties (Wappelhorst & Tietge, 2018).

Electric vehicles are not much better. They are created through high energy use manufacturing which comes at the expense of a massive amount of fossil-fuel use. For some companies, this manufacturing process is 70% higher in greenhouse gas emissions than

¹ This calculation was done by converting 205 GtC by the ratio of CO₂'s molecular weight, so 205 GtC * 3.67 ≈ 752 GT CO₂, then dividing by the current annual emissions rate which is 37.6 (IEA Team, 2025). 752/37.6 = 20.

manufacturing a diesel car.² This is due to the lithium-ion battery in all cars, which can only last ten years before needing to be replaced (Atkinson, 2023). Additionally, the electricity used to charge these cars comes from a highly intensive fossil fuel production process. In 43 states in the USA, charging stations are provided electricity through fossil fuels (Ezeoha, 2020).

Our largest emitter of fossil fuels is electricity and heat at 16.7 billion tons of greenhouse gas emissions a year (Ritchie et al., 2022). This includes heating, charging, and operations in building, households, and infrastructure. Current solutions for electricity and heat related emissions address the meta-problems with transport because they require new renewable energy sources like solar, wind, hydro, and nuclear power. Not only are these solutions incredibly expensive, but they also create a massive initial carbon debt from the manufacturing process, operational inefficiencies, environmentally damaging siting requirements, and a need for fossil fuel backup or massive energy storage to ensure reliable operation. Given these persistent challenges to cleaning the supply, the most reliable and impactful strategy to solving climate change is addressing the demand side. By doing what no nation wants to acknowledge as a solution and making conscious sacrifices to reduce our overall electricity and heat consumption, the globe can significantly reduce emissions for free.

But how does a fragmented globe convince individuals and nations alike to reduce fossil fuel usage? If it cannot be achieved through global coordination, then it must be done by continuous exemplifications of these ethical martyr nations who can do it, thrive, and pioneer the global landscape for centuries to come. However, this kind of global pioneer is unlikely to step forward because of the voluntary sacrifices that would have to be made. Governments must prioritize civilian interests, and their citizens would almost certainly dissent for tragedy of the commons reasons. The best we can do to convince civilians, and by extension governments themselves, is to show what would happen if the globe were forced to become more renewable.

By modeling forced adaptation, infrastructural changes that are physically possible versus politically difficult can be identified. To do so, I came up with a thought experiment: what if an isolated nation was forced into moral clarity by necessity? Iceland's isolation and abundance of cheap and clean energy combined with their relative affluence makes them an ideal candidate for this study.

Research Question

All fossil fuels around the globe—including currently stored fuel—disappear. How does Iceland, a self-contained, energy abundant island nation deal with the fallout?

² It should be noted that this criticism of EVs is often critiqued as myopic, since this carbon debt is paid off over its lifetime – although the amount better it is than diesel vehicles drastically depends on where the car is located and how long it lasts (Moseman, 2022).

Objectives

My objectives in studying this question are threefold: First, I want to understand where Iceland is most susceptible to global supply chain failure. Because they are an island, their reliance on imports could cause major problems since proper plans and infrastructure are not put into place. Second, I want to analyze what Iceland might do when there is direct sociological pressure on the island. With global transport grounded, Iceland would be cut off from aid, forcing internal cooperation for survival. Third, I want to use the recovery tactics Iceland might use as an exemplification of how national sustainability tactics could change right now.

Methods

I have two main points of concern with doing this project: my results becoming a sci-fi scenario with unfounded what-ifs and my results being unclear due to the multiplicity of directions this project will go. To prevent the former, I will ensure all of my explorations are backed by sourced data. To prevent the latter, I will structure my analyses into the shape of Maslow's hierarchy of needs, "a motivational theory in psychology comprising a five-tier model of human needs, often depicted as hierarchical levels within a period" (McLeod, 2025).

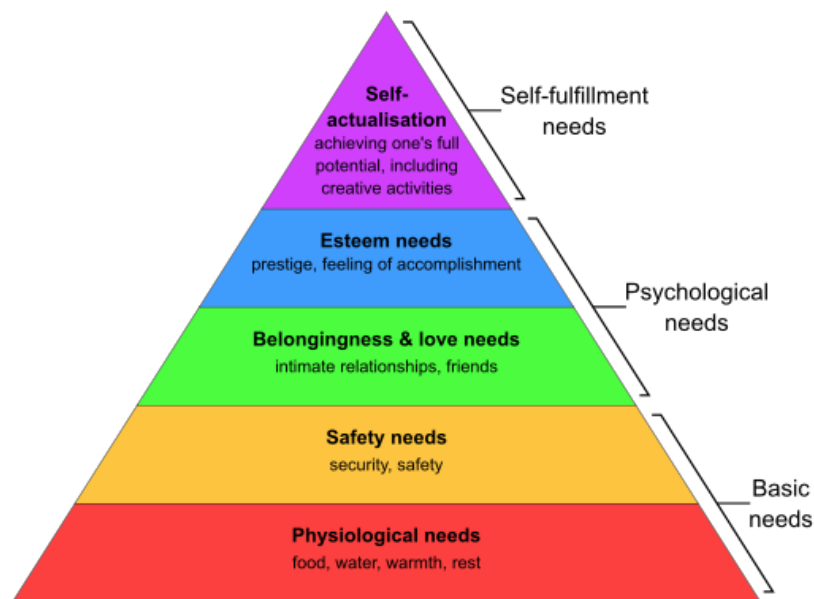


Figure 1: Maslow's hierarchy of needs (Androidmarsexpress, 2010)

Maslow hypothesized that human needs were arranged in a hierarchy, arguing that survival needs must be satisfied before the individual can satisfy higher needs. While more recent studies show that this pyramid is not strictly linear, using this pyramid allows me to deal with each of the public's needs systematically and clearly. This modeling technique is already used in fields like education, healthcare, and workplace management (McLeod, 2025).

Parameters

For this hypothetical, I am instantiating a few parameters and assumptions so that my objectives can be achieved as precisely and in depth as possible. These parameters/assumptions will be an attempt to eliminate extraneous variables that do not aid in the examination of my objectives. With that in mind, the first assumption is that there will be no tourists in the country when fossil fuels disappear. Not only is this an extraneous variable, but it would be difficult to estimate an accurate tourist fluctuation, since foreign presence varies from ~120,000 to ~310,000 depending on month, year, and weather (Statice, 2025b). The second assumption is that all vessels, vehicles, and aircraft are landed when fossil fuel disappears. This is another extraneous variable that doesn't assist in exploring my objectives and is similarly difficult to estimate. Lastly, there could be some confusion on what "fossil fuels" or "currently stored fuel" means. In this study, I will assume fossil fuel extracts/fossil fuel derived fuels (crude oil, natural gas, coal, gasoline, jet fuel, etc.) disappear. Fossil fuel derived products (plastics, asphalt, etc.) do not.

Justification of Methods and Research Question

Since there are various ways my methods and research question could be adjusted to achieve insight into my objectives, I will address potential questions about these alternatives and justify why my chosen phrasing and parameters best serve my aims to explore the objectives.

Why not propose a question where each country gets a five-year set up period before fossil fuels disappear? This question quickly becomes more about international policy, calculations of finite resources distributed around the globe, and near certain full-scale war, instead of about sociological pressure or susceptibility to global fallout. Thus this question only properly addresses my third objective, how Iceland's sustainability tactics could be an exemplification of how national sustainability tactics could change right now.

Why not propose a question where Iceland voluntarily becomes an ethical martyr that makes their entire nation renewable, and give them as much time as they need to set up? This does not answer objective one because it does not force sociological pressure on the society. To make their entire nation sustainable, Iceland would have to move money elsewhere and make their citizens' lives worse. Doing so would make many people leave and would fragment the nation. This also runs into similar problems to EVs. For Iceland to purchase the means to become fully electric and self-contained (since being fully renewable requires them not to purchase produce from non-renewable sources), they would need to purchase or retrofit a massive number of vehicles, vessels, and machinery. This comes with a huge up-front carbon debt that is always being increased by the continual replacement of lithium-ion batteries.

Why use Maslow's hierarchy to model instead of something like a narrative model (i.e., day 1, month 1, month 6, etc.)? While a narrative model is commonly used for scenario situations, it is often used when there are not as many variables. If I were to use the narrative model, I would risk the project becoming messy because each variable would be split up and reiterated after each temporal change. By modeling through Maslow's hierarchy, I can

systematically deal with each issue by salience, effectively crossing each issue on the pyramid off so it can be implemented into less salient issues that might intersect in later levels.

Scope and Limitations

While this thought experiment is grounded in empirical data about Iceland's energy infrastructure, food systems, and resource availability, there are several important constraints that must be acknowledged.

This analysis does not account for the unreliability of individual behavior and compliance with government coordination. In reality, a crisis of this magnitude would likely cause panic, denial, grief, hoarding and resistance to the authoritarian measures outlined below. While I reference Iceland's resistance during the 2009-2011 Pots and Pans Crisis, human behavior under such a physiological disruption of prolonged scarcity is unpredictable and varied.

It will be assumed that the Icelandic government maintains cohesion and operation throughout the crisis as this study does not consider possible internal political fractures, competing factions, the possibility of a coup, or regional communities refusing to comply. It also does not account for potential technical and infrastructural problems, such as the instability of the electrical grid and electric vehicles and equipment malfunctions. Long-term health consequences of dietary restrictions and reduced medical care are not included. The environmental consequences (e.g., overfishing impacts) of relying heavily on domestic methods of food capture are not accounted for.

There are some methodological constraints of the hierarchy framework. While Maslow's hierarchy provides systematic organization, it imposes linearity on what would be a chaotic and simultaneous crisis across all levels. Studies have shown that humans often pursue Maslowian needs concurrently (McLeod, 2025).

Ultimately, this is a speculative exercise. While every claim in the results section is supported by current data, the scenario itself is unprecedented. My analysis suggests what might work based on available resources, but it cannot predict what would work. It is not an exercise in prophetic accuracy, but in systematically thinking about dependencies, vulnerabilities, and possibilities that reveal insights of current sustainability tactics and policy.

Ethics

During my research, I interviewed and emailed professionals sought to discover what was most important to people who are closer to the issue than I am. By the end of my research, I had conducted two interviews: one with the Head of the Environmental Agency of Iceland, Arni Finnson, and another with a Geophysics professor at the University of Iceland, Þróstur Þorsteinsson. In addition to the interviews, I emailed a variety of professionals to confirm my findings. I received a response confirming some of my findings, but due to the sensitive nature of the information they verified, the person did not want their quote or name in this paper, so their response and name were excluded from the project.

To ensure my interviewees rights and needs were considered, I informed them of my research question, my reason for researching the topic, the time frame I wanted to meet with them for, my position as a student in a semester abroad program, and the option to opt out of questions or review more material before the interview. At the beginning of the interview, I began by making sure they were comfortable with me recording, and that if any time they wanted to stop the interview or skip a question to please say so. After the interview, I asked them to sign a consent form allowing me to use their names and responses in my paper. I told them I would amend any rights section and/or exclude information if they so desired.

Artificial Intelligence Statement

AI was used to find specific sources. It was not used for any other purposes in this paper.

Results

Each category of the project will be broken down as follows. Maslow level one: energy, water, heat and shelter, food, and medical needs. Maslow level two: transportation and control. Maslow level three: technology, jobs, and culture. Maslow level four is not addressed because esteem needs are more individual and based on level three needs of belongingness and love. Level five, self-actualization, is even more individual than level four. Since this is an overview of sociological pressures and not psychological pressures, both level four and five are not governmental concerns. The government can only hope that if first three levels are fulfilled, the last two will fall into place of an individual's own fruition. At the beginning of each level, I will justify the positioning of the categories within the level.

Maslow Level One: Physiological Needs

The first level of Maslow's hierarchy contains physiological needs. Energy is not mentioned on Maslow's hierarchy anywhere; however, it is a modern prerequisite for other physiological needs. A nation cannot run their water pumps, heat buildings and houses, produce food, and provide medical support without energy. The rest of the categories on this level—water, heat and shelter, food, and medical needs—are direct physiological needs.

Energy

This is the area where Iceland, by natural availability of resources, thrives the most. They have an abundance of hydropower and geothermal energy, and besides the two island communities Grimsey and Flatey, their nation is entirely on the electric grid (Yu, 2018), which comes from 100% renewable sources (Landsvirkjun, n.d.). Only 9.12% of electricity is used for commercial reasons (services), 4.49% for households, 4.23% for utilities, 1.24% for agriculture, and 0.38% for fishing (Statice, 2018). The rest, 80.54%, is used for aluminum smelting and datacenters. This 80% accounts for 15,525 GWh of power, enough to heat households and power all utilities eight times over. Since aluminum smelters will not have a way to import aluminum ingots or export the made aluminum after fossil fuels disappear, and datacenters become

unnecessary as the globe deals with the collapse, the question becomes not where to obtain enough energy to power the nation, but where to redirect this massive excess of 80.54% of energy.

Water

High-quality freshwater is abundant in Iceland (NSIT, n.d.).

Heat and Shelter

90% of the country is heated with geothermal, 9% with electricity from hydropower, and 1%, mostly in rural areas and small island communities (Flatey and Grimsey), is heated with oil (Samorka, 2025). This means there will be a few small communities that need to move to heated districts. Fortunately, Iceland has an excess of housing due to unoccupied houses meant for tourists (Mbl, 2024).

Food

Iceland's self-sufficiency rate for consumed calories from domestic production is 53% (Meltzer et al., 2024; Table 1). A breakdown of Iceland's imports sheds insight into what products are most imported (Trading Economics, 2024). Many products like beverages, spirits, and vinegar (\$113.50M imported) and animal, vegetable fats and oils (\$100.59M imported—this one can be produced domestically with butter, fish, and lamb) are not vital to their food intake. The largest and most critical imports are cereal, flour, starch, and milk preparation products, totaling \$109.47M. This is followed by Edible fruits and nuts at \$78.41M (Trading Economics, 2024).

Long term, Iceland will have to figure out how to build infrastructure to make up this 47% deficit, but they need an immediate plan to deal with the crisis while they figure out other options. Thus I will break down this category into two sections: immediate action and long term plans.

Immediate Action. Since Iceland has no emergency food supplies (Adamsdóttir, 2025), they would be working with whatever commercial inventory is on hand. The immediate strategy I have outlined will work for a period of several weeks to a few months.

After the 2009-2011 Pots and Pans Revolution that followed the 2008 economic crisis, legislation was put into effect that allows the government to confiscate food, fuel, spare parts, pharmaceuticals, and other vital supplies, and issue instructions about the sale and distribution of necessities if the public so requires (Jóhannsson, 2011, p. 98). Using this legislation, the government would immediately enact a national food security emergency act. They would use their police force to seize control over all food repositories in the nation. A focus would be on the seizure of all bulk stores of imported, non-perishable, high calorie foods like flour, rice, pasta, sugar, cooking oil, and canned goods.

All fish landings would be nationalized, and any food from near-shore fishing would be closely monitored and directed to the national rationing system. The four existing electric vessels would be seized. These include the ferry to Vestmannaeyjar, Herjólfur (Rarik Team, 2020), two whale watching boats in Húsavík, the Andvari (North Sailing Team, 2017) and the Opal (Schooner Team, 2022), and the hybrid research vessel Þórunn Þórðardóttir HF300 based in Hafnarfjörður (Marine and Freshwater Research Institute, 2024).

Since aquaculture would cease to exist due to the farms' reliance on diesel fueled feed barges and specialized service boats that keep out diseases and parasites (Dánielsdóttir, 2021), all fish would start dying on day one or two without maintenance. There would be a small window of opportunity to take all fish out of farms and mass dry and freeze them.

The fleet of four electric vessels would immediately pull all fish out of fish farms in the Westfjords and Eastfjords. Using any small craft available, teams of people would physically net, scoop, or haul fish from the cages. Sailboats and barges could potentially be towed short distances by the electric boats. All fish would immediately be put on ice, then processed and either frozen or smoked.

Simultaneously, all stored animal feed in the country would be seized and audited. An inventory of the quantity and type of feed will be determined, and pigs and poultry will have to be slaughtered due to their reliance on imported, grain-based feed (Jóhannesson, 2010). This slaughter will include 2,984 pigs and 192,245 hens (Statice, 2023c). This meat will either be frozen or smoked. Cattle and sheep will then become the priority for the remaining domestic hay and silage. Dairy cattle will receive the highest priority.

A calculation on the hay deficit will be made for the upcoming winter to inform necessary herd reductions. While larger culls might happen later, Iceland must start to figure out which animals are the least productive or in poor condition to begin killing them as they prepare for the winter reduction. Teams of people will be assigned to cutting hay with a scythe, drying, gathering, and storing hay to obtain as much hay as possible for the winter.

Once all goods are seized, an audit will be conducted to get a precise count of total available calories, broken down by macronutrient (carbohydrates, fats, proteins). This audit would become the basis for their rationing system.

Using the audit, they would set up a food distribution system through existing grocery stores and dole out rations on a per-person, per-day caloric basis. An Icelandic's average daily intake is 2,044 kcal per day (Directorate of Health, 2022). By multiplying this by their population of 389,444 (Statice, 2025a), the number of kcals rationed per day can be acquired: 796,023,536 kcal per day are needed to keep everyone at the same sustenance rate they are at pre-lack of fossil fuels.

However, it is highly unlikely the government would be able to procure and sustain this much food for rations, considering fishing, imports, and farming as Icelanders know it will be drastically changed and/or obsolete. Estimates for a survivable number of kcals a day always have some variability, but it is generally agreed upon that people doing hard labor can survive – albeit struggle greatly - 1500 kcals a day and be psychologically stable with 1900 kcals a day

(Institute of Medicine (US) Committee on International Nutrition et al., 1995). Because most of the nation will be unemployed, 1500 can be a general baseline ration amount. There will have to be a differential rationing system because some people will be burning more kcals than others – construction workers, fisherman, scythers, policemen.³ This will not be a majority of people. Accounting for the differential rationing system, a reasonable estimate an average of 1750 kcals a day can be made. Multiplying this by the population comes out to 681,527,000 kcals/day.

Next, the weight of the food must be calculated to determine how many trucks/vans can transport it around the country. The weight of food is variable, but mixed rations of carbohydrates like flour, rice, and pasta with proteins like fish and dairy averages 2,320 kcal per kg. I calculated this using Nutritionix, the world's largest verified nutrition database that gets its data from automated data collection and manual verification by registered dietitians, to obtain kcal per kg of primary foods in Iceland, and then found the averaged the median and the mean. Proper distribution here is difficult to discern, so I averaged all products the same.⁴ In reality, more wet domestic foods would most likely be used, so this number might be even lower. Rations would blend the scarce imported carbohydrates with domestically produced proteins, dairy, and vegetables depending on current supply.⁵

Long Term Plans. While products like cereal grain and potatoes are grown in small amounts around Iceland (Statice, 2024a), it would barely make a dent on the food deficit. The annual harvest of these two products is 10,623 tons a year, which after converting these numbers to kcals by numbers from Nutritionix by total kgs produced a year⁶, I get ~22,524,290,000 kcal per year. Dividing this by the daily kcal need of Icelanders (681,527,000), one can see the entire annual stock will last only 33.05 days. Additionally, all tractors in Iceland are diesel (Ministry of the Environment, Energy and Climate, 2020), so expansion efforts would be nearly impossible and/or incredibly inefficient to closing the gap. Therefore, the key to closing this deficit is using Iceland's massive energy capacity combined with domestically obtainable products.

³ Addressed in Maslow level three, under jobs.

⁴ Wet domestic food: plain skyr ~610 kcal/kg, raw Atlantic cod ~820 kcal/kg, cooked cod ~1,050 kcal/kg, lamb ~2,100 kcal/kg.

Dry imported food: all-purpose flour/rice: ~3,650 kcal/kg, fats and oils: ~8,840 kcal/kg

Mean: ~2,590 kcal/kg

Median: ~2,050 kcal/kg

Average: ~2,320 kcal/kg

$681,527,000 / 2,320 = \sim 293,761.63$.

⁵ For methods of transporting this food, see Maslow level two, under transportation.

⁶ The total annual harvest of cereals and potatoes amounts to 10,623 tonnes (10,623,000 kg) (Statice, 2024a). Using Nutritionix energy values, potatoes (~800 kcal/kg × 5,514,000 kg) provide 4,411,200,000 kcal, while cereals (an average of ~3,540 kcal/kg for barley, ~3,890 kcal/kg for oats, and ~3,325 kcal/kg for wheat and rye, giving a mean of 3,545.33 kcal/kg) contribute 18,113,090,000 kcal. In total, this equates to 22,524,290,000 kcal per year.

VAXA. This is a company integrated with one of the largest geothermal plants in Iceland, the Hellisheiði Geothermal Power Station. It uses the CO₂ waste output, energy, and cold and hot water from the plant to produce microalgae. These microalgae are produced in a sealed, indoor, bio-secured facility without the use of pesticides or antibiotics. Microalgae is rich in protein and unlike other plant-based sources such as soy, it contains a complete amino acid profile and is rich with phyco nutrients. It is already being used as a meat replacement and alternative protein (VAXA Team, 2020).

The microalgae VAXA uses is called Icelandic Ultra Spirulina. It only needs red and blue wavelengths for photosynthesis, so any white lights will work for it. Its entire system, including lighting, environmental controls, and nutrient delivery, is automated, and is done through a TIA portal. A Siemens team supported VAXA with offline data models that allow them to monitor and regulate every aspect of production, from growth conditions to harvesting (Visser, 2025).

The widely accepted caloric value for dried Spirulina powder is ~2,900 kcal/kg (Lympaki et al., 2022, Table 3). Currently, VAXA grows 120 metric tons of algae per year, but at their one location, they have the capacity for as much as 20,000 metric tons of production a year (Peters, 2022). If they were to produce this amount, they would make 58 billion kcals of microalgae a year. On a per day basis, this is 158,904,110 kcal. Since the total deficit 320,317,690 a day (based on the 47% imported), these 20 metric tons would make up 49.6% of the deficit with just one plant.

While other geothermal plants could be used in expansion efforts, the area VAXA is already located at – the Hellisheiði powerplant - provides the most optimal set up, because it produces by far the most CO₂ at 43,200 tons (Gunnarsson et al., 2013, Figure 3), is close to the aluminum smelter, Rio Tinto ISAL smelter in Hafnarfjörður, and is close to another powerplant, Nesjavellir, which produces 18,600 tons of CO₂ (Gunnarsson et al., 2013, Figure 3) and has a pilot carbon capture project that is already operational (CarbFix Media, 2023).

The Rio Tinto smelter alone produces 202,000 metric tons of aluminum per year (ISAL, 2024). Due to logistical buffering and the fact that Iceland must have all ingots shipped in on boats, it can reasonably be assumed that smelters typically maintain inventory for 2 to 4 weeks. By dividing 202,000 by 12, a monthly inventory of 16,833.33 metric tons can be estimated. This aluminum is more than enough to scale VAXA to their capacity.

Other materials could be sourced from bottling companies, large diesel vessels, and existing infrastructure. While a complex undertaking requiring significant national effort would be required to source and adapt existing resources, scaling this up could prove domestic technology offers a tangible long-term vision for enhancing Iceland's food security until other self-contained sustenance methods can be set up.

The microalgae require chemical feed of Nitrogen and Phosphorous, but since imports are gone, Iceland must manufacture its own nutrient supply for the microalgae. This can be done by converting pathogenic waste into nutrients via composting. The national sewage waste will be diverted from the ocean pipeline, screened with solids removed, and trucked or piped to the VAXA complex. It will be fed through a repurposed industrial geothermal heat exchanger, which

kills pathogens. The nutrient-rich water is then fed directly into VAXA's bioreactors. Animal and fish offal will be sent to the protein hydrolysis plants that already exist in for fish processing around the country. These locations will then process the offal and transport it by trucks or pipes to VAXA. Both strategies create a sustainable method of using domestic products to feed the microalgae.

Fishing. VAXA microalgae would be used for onshore aquaculture farms of Arctic Char. In 2023, 5,248 Char were farmed (2023a) in industrial facilities with fish tanks and water circulation, filtration, oxygenation, and temperature control systems all run by electricity. This number could scale up and make up a substantial amount of Icelander's diet. Additionally, more of these farms could be built inside datacenters or aluminum factories using excess aluminum.

Wild caught fishing using boats like Herjólfur to drop teams of people in small rowboats and kayaks would be set up. On shore fishing and netting would become a vital part of Icelandic diet, and passenger EVs would set up as many of these netted areas in as many fjords as they could to maximize fish intake.

There would also be low-energy sea farming by establishing and expanding mussel, oyster, and seaweed/kelp farms in fjords using small sailboats, rowboats, or the electrics for harvesting. Processing would use grid electricity. Farming these sources utilizes marine space sustainability, adds dietary vitamins and minerals, and provides alternative protein/food sources.

Greenhouses. Iceland is a leading innovator in greenhouses. Their electricity makes greenhouses easy and cheap to maintain, and as of 2012, there are 194,000 m² of total surface area (Butrico & Kaplan, 2018). If expansion is to be conservatively assumed, one can estimate a 25% increase in the past 13 years. This would make the total surface area 242,500 m². While they cannot expand space because almost all parts for greenhouses are imported, they can continue growing using offal/waste fertilizer given to VAXA. Since vegetables do not make a caloric dent, and based on growing area, growing something like hydroponic potatoes will not make a dent either, greenhouse infrastructure must be utilized in different ways.

All existing tomato, bell pepper, and cucumber operations would be maintained, but no longer as a base food. They would be harvested and processed into pulp or juice and added to rations to supplement vitamin C. Since fruit imports are gone, this will prevent scurvy. Lettuce, basil, mint, and thyme cultivation will be massively expanded to make up for the lack of fiber. These herbs can be turned into teas, but also provide a psychological value of flavor to maintain social cohesion. A large section of the greenhouses will be cleared for mass-scale Yarrow cultivation. It will become medically available for those in need. Willow will be grown for their Salicin, which is a natural precursor to aspirin. Lastly, opium poppy will be farmed for morphine, used for emergencies and surgical medicine.

Medical Needs

The medical situation in Iceland is a mixed bag. Like all homes, utilities, and services, commercial or otherwise, hospitals run on 100% electricity, so equipment, utilities, and facilities, will run like normal. However, these hospitals only have backup diesel generators to combat unscheduled grid failures. This means that consistent power is of the utmost salience, and failures will cause death.

The major problem with medicine itself in Iceland is all pharmaceuticals are imported. In 2024, Iceland imported \$267M worth of pharmaceutical goods (Trading Economics Team, 2025). While there are some herbal remedies local to Iceland, they are not regulated or accepted by modern medicine. When imports go away, the unfortunate reality is that there will be unavoidable losses due to the missing pharmaceuticals.

There is no database showing Iceland's current or average stock of medicine. Since there is no emergency plan in action, the best assumption one can make is that Iceland's pharmaceuticals are shipped in as needed. After the COVID-19 pandemic, light was shed on the preparedness of the EU for health emergencies. Because of this Iceland joined an EU joint action to strengthen strategic stockpiles of medical countermeasures. However, it is likely these measures have not been implemented yet, since this joint action was announced in September of 2025 (The Directorate of Health, 2025). Still, it can be conservatively assumed there is three weeks of product already in Iceland. The government will seize all stores of medicine immediately to dole out immediate need.

These three weeks become a critical period where Iceland must begin producing the medicine they can. As noted in the section above, Iceland will start growing willow, yarrow, and opium in its greenhouses so that they can have some store of medicine. Opium poppy is particularly important for medical procedures.

Iceland has a world-class biotechnology sector that can be utilized if materials can be sourced. Facilities like Alvotech's state-of-the-art manufacturing plant, a 13,000 m² facility with 16,000L of bioreactor capacity located in the University of Iceland's science park which previously focused on the production of monoclonal antibodies (Pharmaceutical Technology, 2016) can be used to produce insulin using microbes like *E. coli* or yeast. It is likely that they have both of these materials on hand since producing insulin is a commonly taught subject in university biotechnology programs like the Applied biotechnology master's program currently taught at the University of Iceland (University of Iceland Biotech, n.d.).

As of 2021, there are 20,100 cases of diabetes in adults in Iceland (International Diabetes Federation, 2021), but not all people with type 2 Diabetes need insulin. 1400 people in Iceland live with type 1 Diabetes (Type 1 Diabetes Index, n.d.). A conservative estimate of the type 2 diabetics that need insulin would be 20% (MedTech Europe, 2022), so that is 3,740 people. This gives an estimate of 5,140 insulin dependent citizens nationwide.

The average daily insulin use in 2016 was 62 units (Fuglesten Biniek & Johnson, 2019). Multiplied by 365, we get a yearly dose of 22,630 units/year. One international unit of insulin is 0.0347 mg of pure crystalline insulin (Qu et al., 2018). This means the total pure crystalline

insulin needed per citizen per year is 0.785 grams. Thus the total amount needed to sustain all Icelandic citizens is 4034.9 grams per year, or 11.05 grams per day.⁷

Assuming a three-week buffer period, Iceland needs to cover the need before then. It takes 60 hours to propagate the most common insulin producer, *Pichia Pastoris* (A. Moreno-Cid & Verdú, n.d.). This is the period before it can be put into the tanks. Then, it takes 131 hours to ferment the *Pichia* (Anasontzis et al., 2014; table 4). The purification process is difficult to estimate. Sources say it can take more than three days (Vassilyeva et al., 2017), but since it is a first batch, I will conservatively estimate a ten-day period is needed. This means that the entire process will take 15.46 days.⁸ *Pichia* yields can be as high as 20 g per L (Ahmad et al., 2014). Since the Alvotech bioreactor is 16,000 liters, an extremely conservative estimate of 1 g per L would still cover the entire population's insulin for almost four years.⁹ Not only is the three-week buffer period beat, but excess is so great that they can use the bioreactor for other resources like food production until more insulin is needed again.

While insulin is a major win for the country, there is much work to do in the medical department. The government will have to commandeer EV Vans and equip them with as much medical equipment as they can. They will transport doctors to all towns as help is needed. These vans will be run by trained EMTs in the country and will be taken to hospitals run by trained doctors.

There will be new job openings for people to construct gauzes out of moss and splints out of wood. This will include entire fleets of people dedicated to creating new ways to supply medical needs. This medical ingenuity was used pre-imports in Iceland, with locals utilizing Yarrow to pack wounds and aid healing, Icelandic moss to boil into tea to treat respiratory and digestive ailments, and scurvygrass to treat vitamin C during long winters. It should also be noted that Iceland still has access to the internet, and thus has a massive catalog and database of knowledge to pull from to help them find optimal solutions.

Maslow Level Two: Safety Needs

The second level of Maslow's hierarchy contains safety needs. Feelings of safety are often acquired through consistent security over time. For this project, the two categories are transportation and control. Transportation allows for consistent transport of goods and resources. If the nation knows doctors can reach them, understands they are not stranded in their community, and that base needs like foods and supplies are able to be delivered, they will begin to feel secure and safe over time. By using government control tactics, the nation will feel secure from crime, uprisings, and feel a general contentedness with the 'big brother,' that always has their back.

⁷ $22,630 \times 0.0347 = 785.3$ mg/year. $785.3/1000 = 0.785$ grams/year. $5,140 \times 0.785 = 4,034.9$ grams/year.

$4,034.9/365 = 11.05$ grams/day.

⁸ $60 + 131 + 240 = 371$. $371/24 = 15.458$.

⁹ 4,034.9 grams per year are needed.

Transportation

On land vehicles are one of Iceland's largest dependencies on fossil fuels. Only 7.02% of the nation's car fleet is electric, with a total of 25,238 electric vehicles out of 359,400. 23,601 of these vehicles are passenger EVs, 1,592 are vans, and 45 are heavy rig trucks (Directorate-General for Mobility and Transport, 2025).

The government must immediately seize all EVs in the nation. These vehicles serve as vital components of distribution and order. The truck rigs would be assigned to distributing food across the nation. Each food transport rig would be equipped with electric powered refrigeration – something already supplied in Iceland by companies like Carrier Digi and Thermo King.

The nation needs to move 293,761.63 kg of food every single day. Using a conservative payload estimate of 20,000 kg per electric rig (*Volvo FH Electric*, n.d., this is estimated after weight of truck and trailer is accounted for) and dividing total food needing to be moved by this payload estimate¹⁰, I found that about 15 trips would be needed each day. Since there are 45 heavy rig trucks, there are more than enough to account for charging and shipping time, and still have left over rigs. Also, as of 2024, 63% of the population lives in the greater Reykjavik area (Statrice, 2024b), so most rigs would not be moving that far. Only 37% would be traveling out of this area, so only about six¹¹ would have longer trips.

Vans would be retrofitted as best as possible to support medical teams, and sent out to hospitals proportional to the communities' population. All passenger EVs would either become taxis or part of the police force.¹²

Control

The crucial difference between the pots and pans crisis and this one, is the government is not as much at fault. While they didn't have proper emergency food supplies, the nation will have hope and trust because they understand Iceland has a leg-up from countries like America, China, India, and most of the EU. They have unending energy, and this is how the government will frame the crisis. The nation must come together using their unique gift of geothermal and hydropower energy to fight against the chaos and scarcity to follow. To not capitalize on this gift would squander their chance to save the human race.

Governmental propaganda communicated via the still-functioning internet, radio, and television would be nonstop. They would frame the crisis not as a loss or an end all be all, but as a test. They would speak about the resilience of the nation pre-fossil fuels and call for national unity. The government cannot try to be loved. They must avoid being hated and do so in any way possible. There must be a public-facing tone of regretful control where they ensure that their actions of seizing food, nationalizing labor, and rationing are not a power grab, but an unfortunate burden shouldered for their people.

¹⁰ 293,761.3 kg/20,000 kg $\approx$ 14.69.

¹¹ 37% of 15 trips.

¹² For both, see Maslow level two, control.

Propaganda must do its absolute best to show fairness. Government officials shown in line for rations and any hoarding, especially by the political elite or the enforcement class would receive extremely harsh and public treatment. It may even be beneficial to stage an elite member hoarding and publicly show the consequences.

All public successes must be publicized and commended. The VAXA plants being built will be livestreamed, hospital care publicized, rationing systems shown. Every caloric increase must be coveted as a major win.

Allowing some levels of individual resource gathering will be a major win for the individual. Encouraging individual fishing and berry gathering, and providing baskets, rods, nets, and gathering tools will allow for a better transition from the pre-capitalism to the post-crisis-communistic world citizens must live in. The government will allow travel with the seized passenger EVs by instantiating a taxi system where citizens can state a travel reason through an application system.¹³

The national police and coast guard would be fully consolidated into one single force, where the primary mandate is resource security. They will guard food stockpiles, seed banks, greenhouses, and VAXA facilities. All food distribution convoys (the 15 daily rigs) will be escorted to prevent hijacking. A task force will actively hunt down and publicly punish individuals attempting to hoard food and/or create illegal markets for this food. Another force would be in charge of ensuring assigned labor corps report for duty at the hay fields, fishing nets, or construction sites.

Maslow Level Three: Belongingness and Love Needs

The third level of Maslow's hierarchy contains belongingness and love needs. The included categories, technology, jobs, and culture all address some aspect of this level, and often intersect with level four (esteem) needs. Technology is one of the easiest modern ways of connecting people. They can communicate with others, understand what the government is doing, and share common interests throughout the country. Jobs create a sense of esteem and connectedness to others in the nation. When people are working to keep everyone else alive, a fundamental feeling of intimacy and love for their fellow citizens is felt. Culture intersects with almost every level because it gives people common goals and shared interests.

Technology

When fossil fuels disappear, all datacenters will kick into a UPS (uninterruptible power supply) battery that gives the datacenter 10-15 minutes before it dies completely (Avelar, n.d.). This gives Iceland an extremely short window of time to download necessities. They would immediately seize 100% of the nation's bandwidth, and begin downloading technical manuals, blueprints, and firmware for their own infrastructure. These are the essentials. Next, they download complete mirrors of Wikipedia, medical databases, agricultural and aquaculture

¹³ See Maslow level three, technology.

textbooks, and engineering libraries. This is for future knowledge. Then, they download the software to use this data: Linux, database software, coding libraries, and OS installers. Lastly, they grab libraries of music, literature, film, and art. This is for sanity and belonging. Everything else they download is a bonus – all types of installers, video game databases, reddit, YouTube, social media code, etc. It should be noted that people in Iceland will have some of their own materials downloaded, which could become a fantastic way of connecting with others and sharing in their community, thus creating more belonging. Once the servers go dark, the governments will put this information in a new national archive, using some of the facilities from the now obsolete datacenters in the country.

RUV would become a vitally important part of Icelanders' life, because it would keep people updated on the goings-on, help further culture, and become the government's main method of controlling citizens. A national logistics registry where the ~25,000 passenger EVs can be dispatched for travel would be made public, allowing people to fill out a form to travel from place to place. These people would be driven by nationalized taxi drivers, and approval would become a government sector in its own. A government assignment board for laborers would be made, tracking every citizen's skill and assigning them to critical tasks.

Lastly, the government would seize all existing tech inventory from stores. This would become a rationing tool for critical personnel. A repair team would strip thousands of decommissioned datacenter PCs for their valuable parts. This will keep ventilators going, be used for grid controls, and help with VAXA automation.

Jobs

Mass employment is a belonging crisis in itself. When people do not have anything to do or work towards, mental health problems arise (Pappas, 2020). Thus jobs are not just important for keeping the country running, but for deep psychological and sociological reasons.

The job market will be replaced by a national labor board. Using the government assignment board, the national labor team would assign people based on skills and incentivize them by giving them extra calories. This not only makes up for the deficit these laborers have for working long days but is also a status symbol and morale booster. These people need to feel like heroes, or they will give up and not help the nation become better.

There will be three different classes of rationing. The highest calorie group will be the people making the missing calories. These are the new elite, and ought to receive 2,500-3,000 kcal/day. This includes the VAXA team – engineers, construction workers, and biologists, the fishing team – on Herjolfur, the Andvari, and the Opal, plus the onshore net-tenders and fjord-farmers, and the farming team who have possibly the most grueling job – the dairy farmers, sheep tenders, and national hay-scything teams.

The second highest calorie group will be the arterial group who keep the system moving. They get 2,000 – 2,200 kcal/day. This includes the logistics team – the 45 heavy rig drivers and 1,592 van drivers moving food and other components, the repair team – electrical engineers,

welders, and technicians, and the medical team – doctors, EMTs, and the new herbalists/splint-makers.

The lowest and most common group will be on the base ration, who receive 1,750 kcal/day. This includes everyone, but also the people managing the society itself. It is vital that these people get base rations to ensure no corruption is going on and instill equality and sanctity into the citizens. This includes the police force, the people running rationing centers, managing the labor board and operating the travel-permit website, taxi drivers, teachers, RÚV broadcasters, and archivists.

Culture

Culture is vitally important to post-crisis communities living within a near-communistic state. When culture dies, becomes suppressed, becomes a hollow meaningless ritual, or is not actively held up by the government, society collapses (Yurchak, 2006). It is not only salient that culture becomes strong and impenetrable, but that it becomes this way through shared meaning. This meaning can no longer be the capitalistic goal of money, or an anarchical pursuit of individual means, but must become a collective fortitude, a resistance against the chaotic forces pressing down on the nation. The nation of Iceland no longer comes together through a Lockean contract of desire, nor through a goal of maintaining the status quo of the past—Iceland comes together through a fight against the finite born out of love for their neighbors.

This is accomplished by focusing on two things: the finite past and the communal present. The national library of art becomes the new cultural canon. RÚV would broadcast a downloaded movie every day, heated public spaces like swimming pools, libraries, and art centers would become places of discussion and togetherness. People can listen to music, read, or talk together. Private media already downloaded becomes a priceless social tool. A person's downloaded folder is a great asset for making friends and building community. Sharing their finitude becomes a beautiful act of kindness.

Status no longer comes from symbols, but from usefulness. The repair team will bleed into every facet of society. Mending, fixing, and maintaining everything becomes a point of national pride – something already practiced in Iceland today through reusability and sustainability practices (Rest in the West Team, n.d.). A local renaissance is born. Local crafts explode. Knitting circles, woodworking, metalsmithing with scavenged aluminum becomes a favorite pastime. The national propaganda RÚV tells will instantiate new heroes who fix problems in ingenious ways, help the community, and find more ways to help society. Every accomplishment is lauded with national headlines and honors.

Even with survivable rations, people will be hungry. When you cannot control how much food you get, it makes you hungrier (Baker & Keramidis, 2013). People will deal with this in two ways. Firstly, they will passively watch food documentaries, YouTube, and other places where food cravings can be quenched through visual perception. The second way is an active participation in food gathering. Much of people's free time will be spent fishing – possibly communally so all goods can be shared within the groups – and gathering berries – also

communally. Berry picking would become a huge deal in the summer, and jams and preserves would be made to prepare for the cold, dark winters. Communal gardening will also become a big deal. Because there is no fertilizer, people will likely use their own feces to grow plants in the summer.

Discussion

The preceding results have modeled how Iceland might survive the immediate fallout of a lack of fossil fuels. It is shown that the nation would have to re-prioritize, moving from a stable, globalized economy to an insular, survival-focused state. They would not only have to change their methods of resource distribution and domestic production, but radically reorient their sociological and political norms. To fully understand this reorientation, I will explore my three primary research objectives: where Iceland is most susceptible to global failure, the stark realities of such direct sociological pressure on an island nation, and what the recovery tactics outlined in my results can show about changes in national sustainability tactics currently used around the globe.

Susceptibility to Global Failure

By far and away the most salient lesson learned from this hypothetical is how globally dependent Iceland really is on not just oil, but resources in general. With 47% of their food coming from imports, an emergency food plan is a necessity. Without grains, they are left with a dwindling supply of what is already in stores, with virtually no way of recovering the deficit domestically. There are two paths of resolution for this issue: either they figure out a way to domestically produce grains in such a way that it is profitable, or they store away a massive stockpile of reserves. The former would be incredibly difficult because of their harsh environment. Production could be scaled, but it is not salable or remotely profitable because everything must be shipped elsewhere. In 2023, Iceland exported a mere \$28 of wheat to Spain, while they imported \$15.2M from a multitude of places in Europe (OEC, 2023). Additionally, Iceland's only grain mill shut down earlier in 2025 (Adamsdóttir, 2025). The more reasonable option is the latter, which has already been researched and advocated for. Experts in sustainability at the University of Iceland suggest a bare minimum of a three-day supply but stress the importance of planning for up to six months (Adamsdóttir, 2025). If a major volcano environmental disruption happened, it is possible boats and planes would not be able to get to Iceland, leaving it stranded. The 2010 Eyjafjallajökull volcano eruption is direct evidence of this. The eruption caused 95,000 flights to be cancelled and ash to cover not only Iceland, but places as far away as Scandinavia (Staff & Wires, 2010).

In my interview with a professor of geophysics at the University of Iceland, Professor Þröstur Þorsteinsson provided an insightful outlook on why neither the former nor the latter is a focus for Iceland: "It always comes down to the same thing, you know, economics, money. In the beginning, these things cost money and they are not profitable... we are an island in the North Atlantic, you know, if something happens to transport, we are very isolated, so we really should

think more about it... As soon as you start to emphasize domestic food production, prices are likely to go up a little bit and then there's the pressure in the other direction to import more. So it's this little pendulum that goes back and forth, I think, like so many other things."

Pharmaceuticals are another massive global dependency that could be disrupted if the supply chain breaks down. Medicine is arguably worse than food, since it often requires complex machinery and is done in hyper specific places that have a patent to produce and distribute it. Fortunately, Iceland has joined a new EU joint action to strengthen strategic stockpiles of medical countermeasures. This action was launched on September 9-10, 2025, in Helsinki, Finland, and involves 25 European countries and 54 institutions (The Directorate of Health, 2025). Beyond this, I suggest storing crucial pharmaceuticals for at least a year in advance. Since most critical medicine is taken consistently and can be predicted, this doesn't seem like an unreasonable suggestion.

It is also not unreasonable to suggest the dissolution of diesel land vehicles in Iceland. A nation with so much free and cheap electricity should not rely on diesel for their cars. While the up-front cost of EVs and the implementation of charging infrastructure is costly, it pays out tenfold in the future. This sentiment was echoed in my interview with Professor Þorsteinsson, who argued that electrifying the vehicle fleet is a clear "low-hanging fruit" for Iceland. He expressed confusion that the government hasn't pursued this "inexpensive or even profitable" solution more aggressively, especially given its existing climate policies.

By charging cars instead of using gasoline, individuals save massively in the long run. To implement this change, I suggest a similar strategy to Norway. Ban the sale of diesel and gasoline vehicles (all of Norway), have a congestion tax for non-EVs and increase the number of toll gates/roads with free passage for EVs in Oslo. If a government makes it inconvenient to own a diesel car and far cheaper to own an EV, the nation will adapt (Portvik & Christiansen, 2018). For a nation whose entire population is about 7% the size of Norway's and with 63% of its citizens living in the greater Reykjavik area, it is a no brainer to make the switch.

A less salient angle of the electrification of all vehicles is the need for electrification of at least a few fully electric fishing boats. This project has shown that Iceland's fishing fleet's dependence on oil is enormous. The two small whale touring boats in Húsavík, a short distance ferry to Vestmannaeyjar, and a research vessel would not be able to account for the fish farms if there was a collapse in the global supply chain. Investing in a few electric vessels is not only good for dependency reasons, but if aquaculture farms could become fully electric, they could market their Icelandic fish as 100% renewable. Fully electric barges already exist through a company called Port-Liner. While the price tag is steep – 3.5 million euro for the larger barges and 1.5 million euro for the smaller ones (WCN Editorial, 2018), having some of these in the harbor to ensure domestic food production of fish can always be maintained is vital.

Stark Realities of Sociological Pressure

The unfortunate reality of a crisis of this magnitude is that the pressure created does not result in simple, voluntary, bottom-up cooperation. In a capitalistic world, tragedy of the commons rules over all. People are motivated by money, and money incentivizes individuals capitalizing on any chance to get a leg-up. This can be shown by how markets move. When prices go down in anything (e.g., stocks, housing, vehicles), people jump on the opportunity, hoping to get a leg-up when the market goes back up. However, this thought process relies on an individual's trust in the market, so when an actual crisis happens, two things happen concurrently: people attempt to obtain as many finite resources as possible, hoping to either keep the items or sell/trade them to get a leg-up, and they continue to have a subconscious, long-term trust in the market where rebounds are assumed. This is the modern tragedy of the commons deeply related to level two of Maslow's hierarchy.

Knowing this, the government must penalize this mindset and reorient it to a collectivist who trusts in equal and fair distribution by their authority. Thus, the government must re-write the social contract, enforcing a top-down, highly centralized authoritarian government. As shown in the results, individual liberties – such as private property, free commerce, freedom of career – have to be immediately suspended. The only way this is possible without widespread revolt is to convince the public it is not the government's fault. Fortunately, in this scenario, it is not. The government must phrase this takeover as one of regretful control for the sake of solving an unforeseen crisis.

Therefore, the most critical recovery tactic shown in the results is not technical, but psychological, because with individual compliance, none of the technical solutions could come to fruition. The nation's survival would depend on its ability to be convinced by the unified national narrative the government puts out. The use of a calorie-based class system is the ultimate expansion of this: it is not just a physiological necessity, but a powerful social incentive that transforms survival itself into a reward for cooperation.

Redirecting by Force: A Lesson for Current Sustainability

The most translatable lesson of this project comes from the sociological side of things. As shown above, without a majority of individuals complying, a nation cannot run. This rule applies on the global stage as well because of every nation's reliance on global supply and demand. If a nation is ostracized from trade, they follow similar tactics to an individual ostracized from a society: they look for other places to turn, and if they cannot find places to fulfill their demand, they will attempt to cooperate from the society that ostracized them in the way that is most likely to allow re-inclusion. By using this knowledge, a rule can be made to reduce global emissions by force.

If a group of wealthy nations form a pact where they only trade with nations that fulfill (x) threshold of renewability, then the ostracized members will slowly follow. Since wealthy nations contribute more to the demand side of resources than poor nations do, a pact like this would cause serious disruption for major suppliers like China, India, and Vietnam. These nations

would effectively weaponize their demand side power, creating a non-negotiable norm for the global market, transforming climate compliance from a voluntary, often-neglected goal into a matter of economic survival.

However, a plan like this would almost certainly come with lash back and claims of eco-colonialism. It is likely Asia would come together and work things out, creating a rival economic bloc. This would not solve climate change; it would simply fracture the world. There are other possible solutions that are less destructive. For example, these richer nations could create a multi-trillion-dollar fund used to finance the green transition of the global south. They could offer China/India a deal where they will finance the closure of their 500 most polluting fossil fuel plants and replace them with renewable alternatives in exchange for a binding agreement to meet (x) threshold of renewability by (y) date. They could also use Norway's model and heavily tax/tariff non-green products, incentivizing citizens to buy the cheaper product.

Conclusion

This thought experiment shows that the true problem with the disappearance of fossil fuels is not that a country cannot run, but rather that the collapse of globalization severely hinders a nation's ability to provide ample resources for its citizens. The results demonstrate that survival is technologically plausible, but hinges on Iceland's ability to leverage their massive surplus in renewable energy to solve its greatest weaknesses.

However, the technical pivot is secondary to the sociological one. Survival would be contingent on top-down, authoritarian rewriting of the social contract. This internal, national-level coercion is a metaphor for the global climate dilemma. It shows that our current voluntary, incremental, and market-driven climate solutions are misdiagnosing the problem. The globe is attempting to solve a collective action problem with individual incentives. Ultimately, the Icelandic model forces a confrontation with Gardiner's perfect moral storm. It suggests that a true post-carbon future requires a level of collective action and global governance that is far beyond our current political imagination. It posits that the true barrier is not a lack of green technology, but a lack of political will to implement the forceful changes necessary for our collective survival.

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